

SESM3032: Heat Transfer and Applications

COMSOL FIN ANALYSIS

WILLIAM DEELY (WTD1G21)

Section A: Modelling and Optimisation of Radial Fin

A1: Defining Parameters

This section of the report considers a radial fin, of inner diameter 25mm and rectangular profile, for example that which may be seen on a ‘finned tube’ radiator.



Figure 1: Finned Tube Radiator

Copper is selected as a material due to its high conductivity of $400\text{W/m}^2\text{K}$. This type of radiator surface may be seen in an industrial heat exchanger, or to heat a room where space is restricted. A value of $h = 10\text{W/mK}$ is chosen as a convection coefficient, to simulate low speed air blowing across the radiator surface. For example, a breeze, or a fan blowing in the immediate area. A value of $T_b = 450\text{K}$ is chosen, and $T_\infty = 300\text{K}$ is chosen as approximately room temperature.

A2: Steady State Solutions for Radial and Straight Fins

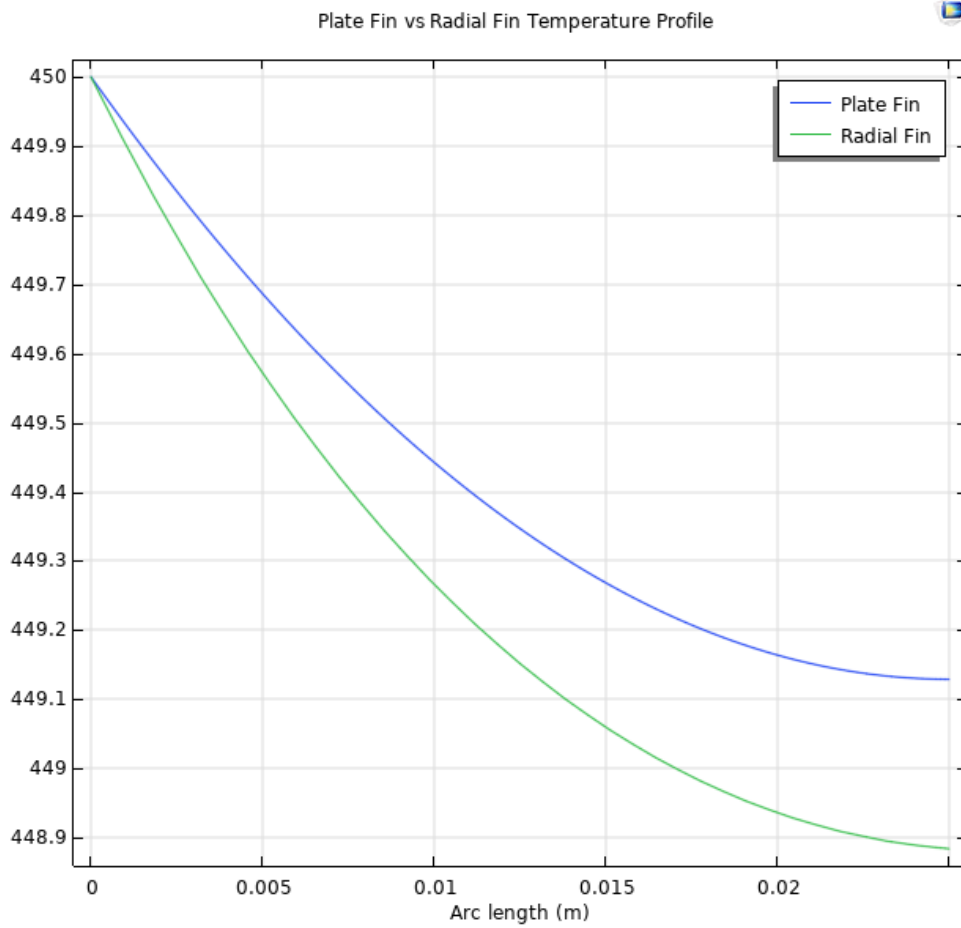
Figure 2 shows the difference between the steady state temperature profiles along the centreline of a 2D axisymmetric section of a radial fin, and a straight plate fin of equivalent length. The dimensions of the radial fin are tabulated below.

Table 1: Dimensions of initial radial and plate fins

Fin Dimension (Section of Radial Fin)	Length (mm)	Fin Dimension (Section of Plate Fin)	Length (mm)
Inner radius, R_i	25mm	Length, L	25mm
Outer radius, R_o	50mm	Width	25mm
Thickness, D	3mm	Thickness, D	3mm

As is shown in figure 2, the radial fin gives a slightly lower tip temperature. This means that it is slightly less efficient than the plate fin, as the ratio between heat transferred by conduction and convection is higher. The radial fin is slightly closer to an “Ideal Fin” of uniform temperature profile $T(x) = T_b$.

Figure 2: Temperature Profile of Centreline for 2D sections of a Plate Fin and a Radial Fin.



A3: Fin Efficiency, Effectiveness and Comparison with Analytical Solution

The efficiency (η) and effectiveness (ϵ) of the radial fin are as follows:

$$\eta = 99.6\%, \epsilon = 26.89$$

Fin parameter, $F_p = mL_c = 0.11449$. Fin parameter may be used in conjunction with the following chart to find the analytical solution for fin efficiency:

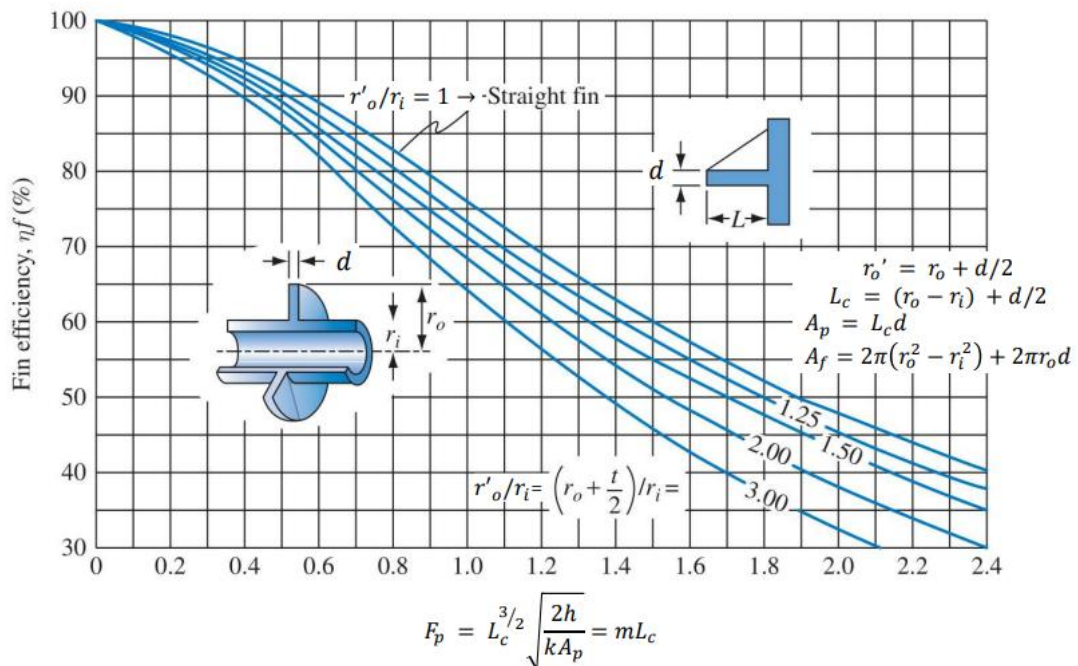


Figure 3: Fin Efficiency vs Fin Parameter, plotted for various ratios of corrected radius to inner radius [1]

For the radial fin design, $\frac{r_o'}{r_i} \approx 2$, so the calculated efficiency in COMSOL corresponds with the analytical solution from the lookup table. This suggests that the model has been defined correctly, and that it is functioning as intended.

A4: Improving the Fin Design

The fin effectiveness for the initial design is low, at 26.89 and efficiency is very high at 99.6%. However, very high efficiency is not the optimal use of material or space. By increasing the length of the fin, and reducing the thickness, the fin transfers much more heat per unit mass and perhaps more importantly, per unit width. Reducing thickness allows more fins across the same width, for much higher total heat transfer. Increasing length increases the fin's surface area, which increases the area available for convective cooling to occur. The performance parameters for the thinner, longer fin are as follows:

Table 2: Radial Fin Characteristics for 2 geometries

Parameter	Initial Fin	New, Improved Fin
Outer Radius, R_o , mm	50	85
Inner Radius, R_i , mm	25	25
Thickness, d , mm	3	1.5
Heat Transferred, $\dot{Q}$, W	18.892	56.71
Efficiency, η , %	99.46	90.01
Effectiveness, ϵ	26.85	240.69

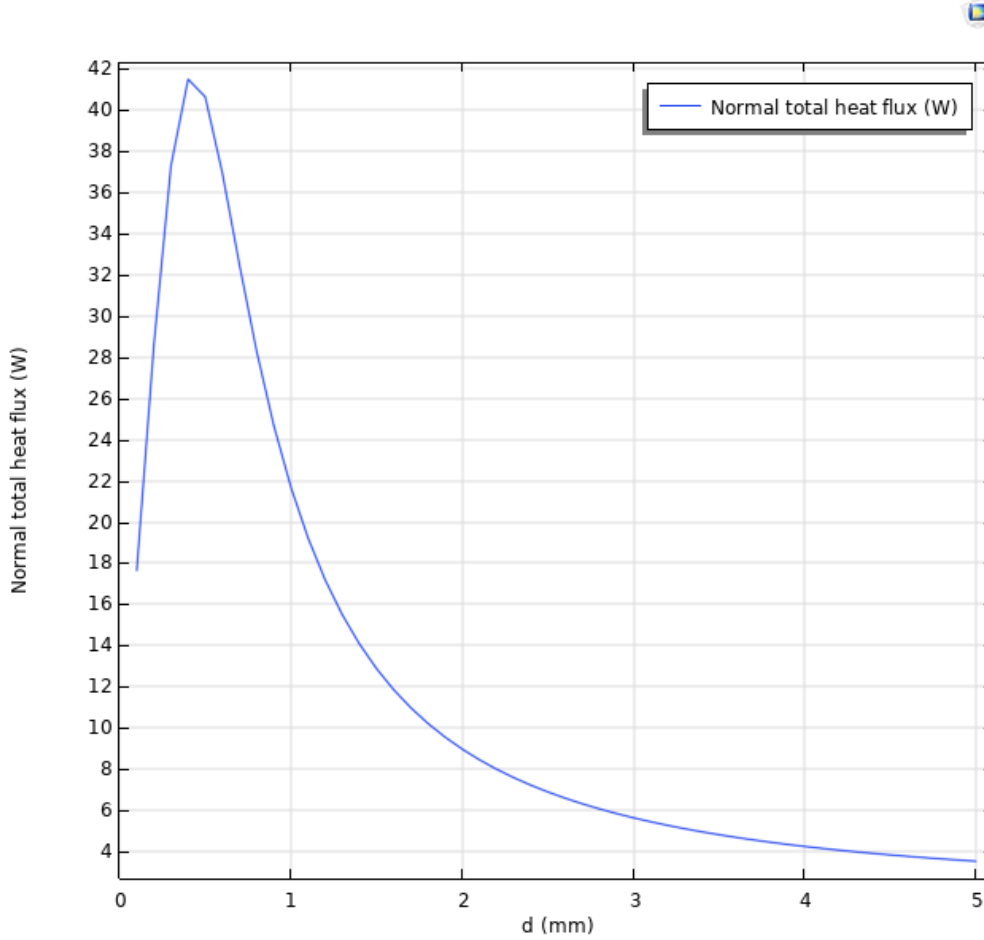
As we can see from Table 2, increasing length by 35mm, and decreasing thickness to 1.5mm enhances heat transfer significantly, and per unit area at the fin root, is almost ten times as effective when compared with heat transfer for the surface without a fin. This effect is more pronounced when an array of fins is used, as an array of the same width can transfer significantly more heat for the improved geometry.

Theoretically, we could change fin thickness to 0.1mm and length to 1m, to further increase effectiveness, as a longer, thinner fin transfers much more heat per unit volume. However, this fin would not be mechanically stable under its own weight. For this reason, thickness is set as $d=1.5\text{mm}$.

A5: Heat Flux as a Function of Fin Root Thickness

Performing a parameter sweep of fin thickness against normal heat flux, while maintaining a fixed profile area, $A_p = 50\text{mm}^2$, by varying outer radius as a function of profile area and root thickness, allows us to select an optimum fin thickness, at which maximum normal heat flux occurs.

Figure 4: Normal Heat Flux as a Function of Root Thickness, d



Optimal fin thickness is 0.4mm which corresponds to a length of:

$$L_{fin} = r_o - r_i = \frac{50\text{mm}^2}{0.4\text{mm}} = 125 \text{ mm}$$

And results in total normal heat transfer of $41.45\text{W} = 263.87\text{W/m}$. (Heat transfer per unit fin width at root)

For a plate fin of profile area $A_p = 50\text{mm}^2$, optimal thickness is given by

$$d_{opt} = \frac{A_p}{L_{opt}}$$

$$L_{opt} = \left(\frac{kA_p}{h} \right)^{\frac{1}{3}}$$

Which gives $d_{opt} = 0.3958\text{mm}$, $L_{opt} = 125.99\text{mm}$. For a plate fin of these dimensions, heat transfer for the given temperature boundary conditions (convection to 300K at the tip, fixed $T=450\text{K}$ at base), heat transfer at steady state is 237.69Wm^{-1}

Heat transferred per unit length is slightly higher for the radial fin (taking length at the outer circumference of the fin). This is likely because of the higher area available to transfer heat for the annulus in comparison to the cuboid for the same root length.

In both cases, the optimal fin geometry is not feasibly manufacturable. A fin, either radial or straight, of length $>100\text{mm}$ and thickness $<0.4\text{mm}$ is unlikely to be able to support its own weight. An array of these fins is likely to deform and have poor performance in comparison to their theoretical, infinitely rigid forms.

As shown in Figure 5 (below), fin root thickness is proportional to fin efficiency, up to a certain limit, where the fin length becomes small enough such that temperature is uniform along the length of the fin (i.e. 100% efficiency). A fin of this profile is highly efficient, however is not effective at removing heat.

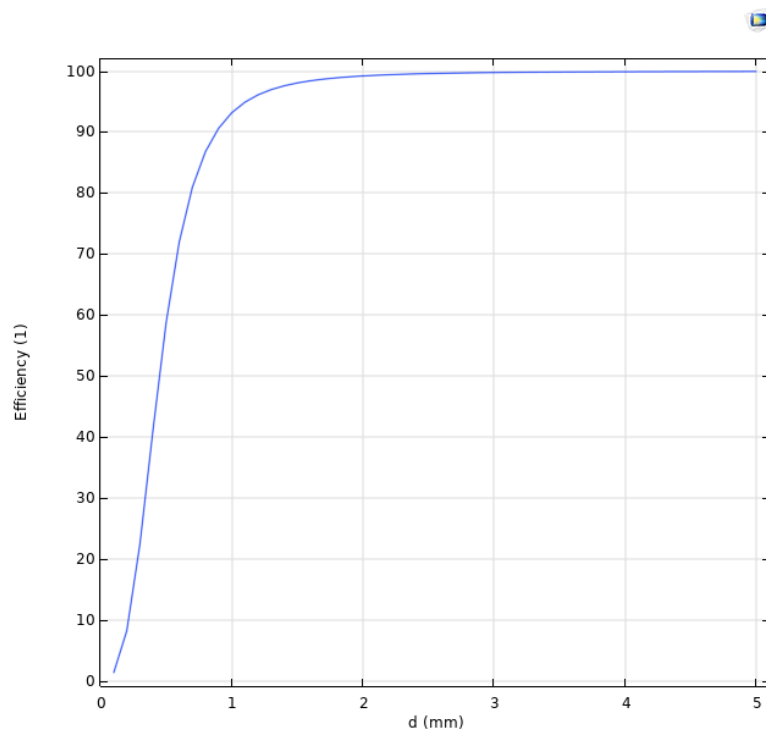


Figure 5: Efficiency as a function of Root Thickness

A6: Efficiency, η , as a function of Fin Parameter, F_p .

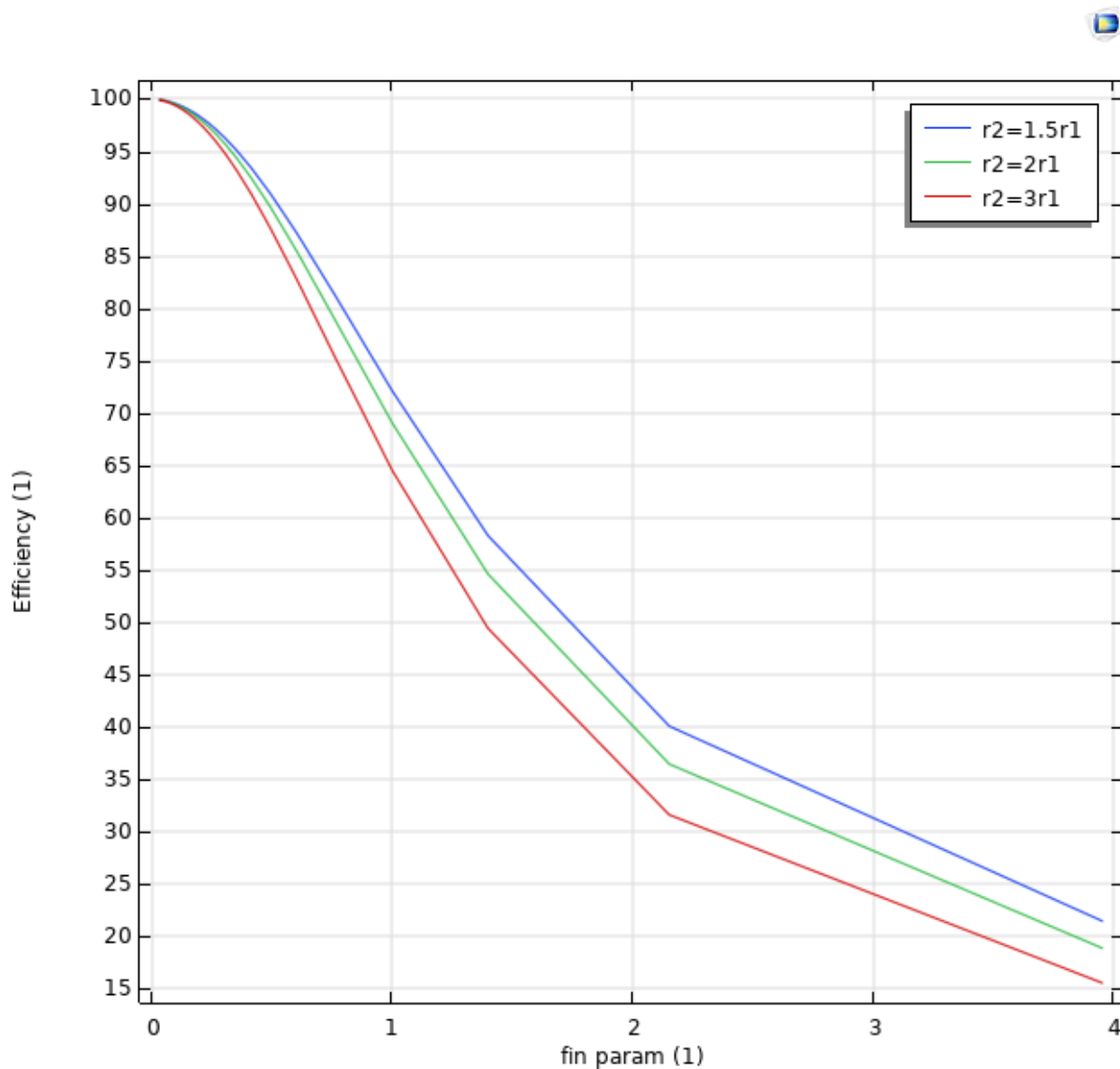


Figure 6: Efficiency as a function of Fin Parameter for three ratios r_2/r_1 for a radial straight fin.

Figure 6 shows the relationship between efficiency and fin parameter for a radial straight fin of various ratios of outer to inner radius. Profile area is fixed at 50mm^2 . This figure shows that a higher ratio between outer and inner radii corresponds to a lower efficiency. However, radial straight fins of a higher ratio are more effective at enhancing heat transfer.

The graph corresponds closely with Figure 3, which demonstrates this relationship analytically.

A7-A9: Radial Fin Array

In this section, a axisymmetric 2D section of an array of 13 radial fins is set out (fin geometry optimised for $A_p = 50\text{mm}^2$) as though it is carrying condensing steam at 100°C . The inner surface of the pipe wall is given a fixed temperature boundary condition of 373.15K (100°C). The outer surfaces of the fin and pipe are given a convective boundary condition, with T_∞ , the ambient temperature, set at 298.15K or 25°C . Convection coefficient is left as $10\text{W}/\text{m}^2\text{K}$ to reflect slow moving air in an air-conditioned industrial setting.

For a radial fin array of length 1300mm , pipe wall thickness= 2mm , fin length= 125mm , inner pipe diameter = 46mm , specific latent heat of condensation= 2257kJ/kg , condensation rate is:

$$\dot{m} = 701.77/2,257\text{kJ/kg} = 0.311\text{ g/s}.$$

$$\dot{m} = \frac{Q}{L_{cond}}$$

Where $\dot{m}$ = mass flow rate of condensed steam, Q = rate of heat transfer, L_{cond} = specific latent heat of condensation.

Reducing the length to 650mm (while maintaining the 13 evenly spaced fins) reduces condensation rate to 0.277g/s, and further to 260mm reduces condensation rate to 0.257g/s. A longer condenser has slightly higher rates of condensation, due to additional heat transfer through the bare pipe surface, however it is more effective to use a smaller fin spacing over a longer pipe.

Figures 8-10 are taken from a “Cut-Line” dataset and show the variation of temperature through the middle of the pipe wall down the length of the pipe, perpendicular to the fins (See Figure 7)

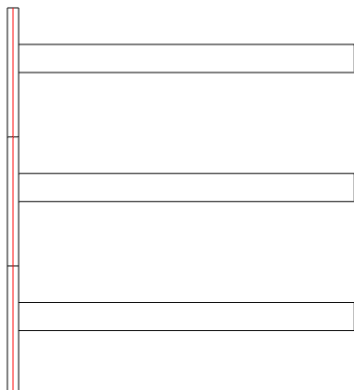


Figure 7: Cut Line Dataset location (Red)

The temperature profiles for each configuration are shown below:

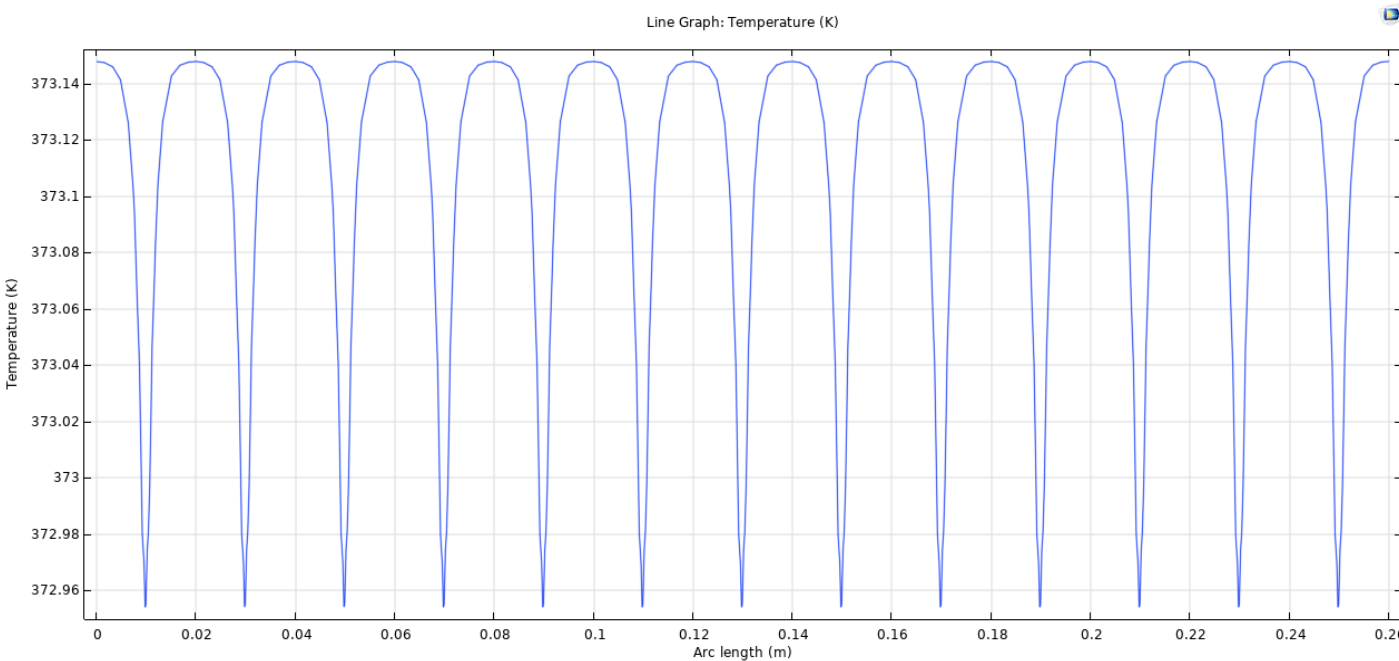


Figure 8: 20mm Spacing

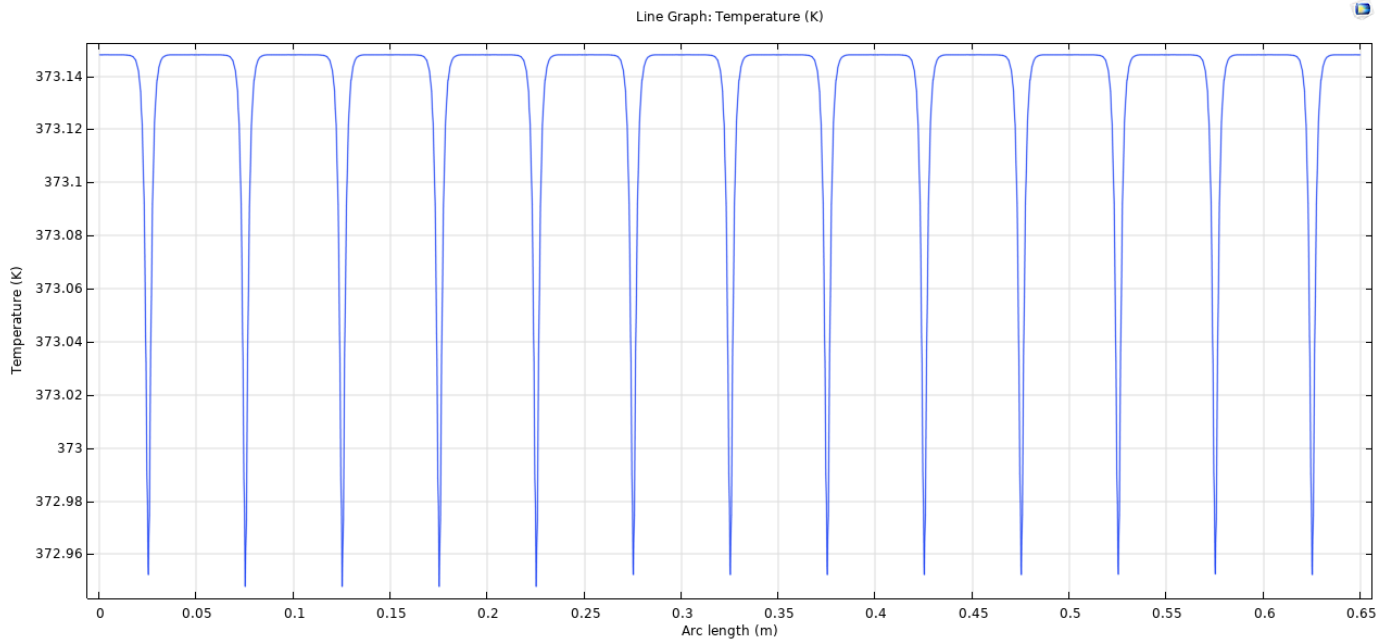


Figure 9: 50mm Spacing

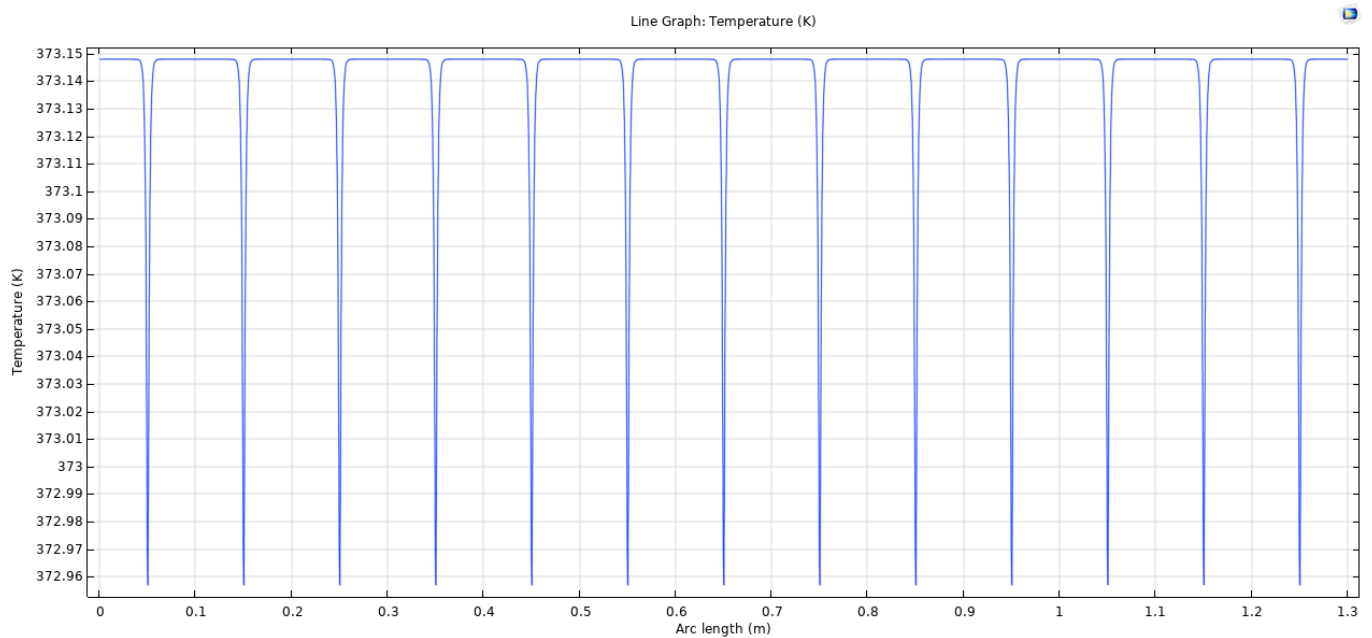


Figure 10: 100mm spacing

The temperature profiles clearly show peaks in temperature at locations which correspond to the fin root (in the case of 100mm spacing between fins, peaks occur every 100mm). This is because the pipe wall outer surface has a convective boundary condition at the surface, whereas at wall locations where there is a fin, the heat is transferred along the fin. The higher temperature at the fin locations shows that the heat is transferred more efficiently through the fins than convectively at the pipe surface. This may also be observed in the revolved 3D plot in figure 11. The low points in temperature occur at the points equidistant between the fins, where the heat transfer regime is dominated by convection.

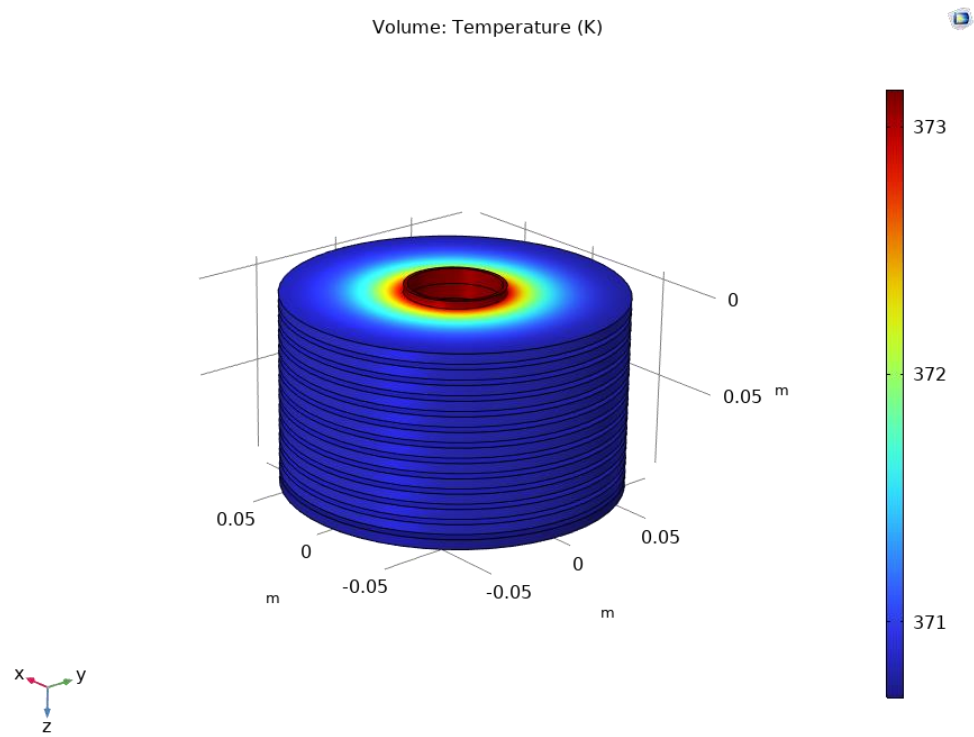


Figure 11: 3D Temperature Plot of Radial Fin Array.

Section B: Design and Modelling a Pin-Fin Array for CPU Cooling.

This section of the report contains heat transfer analysis of heat dissipation for a CPU.

B1: Comparing heat transfer coefficient with and without a heat spreader.

Using Newton's Law of Cooling, required heat transfer convective coefficient for a max temperature of 75°C for a 3mm radius chip with and without a 45mm radius heat spreader is as follows:

Without Spreader:

$$h = \frac{\dot{Q}}{A(T_s - T_\infty)} = \frac{55W}{(3 * 10^{-3})^2 \pi * (55)} = 35,367.78 W/m^2K$$

With Spreader:

$$h = \frac{\dot{Q}}{A(T_s - T_\infty)} = \frac{55W}{(45 * 10^{-3})^2 \pi * (55)} = 157 W/m^2K$$

Without the heat spreader, a very large amount of heat is transferred through the CPU chip. The convective heat transfer coefficient required to cool the chip to 75°C is not feasible to achieve using forced convection of air by a CPU fan. The heat spreader allows the same load to be spread over a much larger area and be dissipated by a convection flow of heat transfer coefficient 157 W/m²K, which is more realistically achievable with forced convection.

Resulting CPU temperatures for 0.5W and 5W for the CPU with and without the heat spreader are shown in Table 3, further proving that the heat spreader plate is necessary.

Table 3: Resultant CPU temps for 0.5W, 5W, with and without heat spreader plate.

Heat Flux (W)	Temperature With Plate (°C)	Temperature Without Plate (°C)
0.5	132.6	1146
5	20.5	25.006

B2: Pin Fin Design

A value of $h = 157.38 W/m^2K$ is derived below for forced convection of air over a cylinder. This is based on a moderately high airflow velocity of 5m/s and the following values:

$$Re = \frac{\rho UL}{\mu} = \frac{1.2 kg/m^3 * 5 * (2 * 10^{-3})}{1.81 * 10^{-5} Pa \cdot S} = 21,116.02$$

$$Pr = \frac{\mu * C_{p,air}}{\kappa} = \frac{1.81 * 10^{-5} Pa \cdot S * 1006 J/kgK}{25.87 * 10^{-3} W/mK} = 0.704$$

$$Nu = 0.683Re^{0.466} * Pr^{0.33} = 12.591^1$$

$$h = Nu * \frac{\kappa}{L_c} = 157.38 W/m^2K$$

¹ Nusselt Number Correlation table available at [3]

A fin volume of 200mm^3 is also selected within the design space of $50 \leq V_f \leq 200\text{mm}^3$. The upper limit is selected for higher surface area, to enhance convection. This gives a length of 63.7 mm for a fixed diameter of 2mm. As the pin fin is likely to be set out in an array, the convection effect is reduced by adjacent fins. Thus, the heat transfer coefficient, h , for the following analysis is scaled down to $100\text{ W/m}^2\text{K}$ as an average value across the array.

B2: Parameter Sweep: Fin Diameter vs Heat Flux

For the following analysis, the following parameters are assumed:

Table 4: Parameters for Pin Fin Diameter analysis

Parameter	Value
κ_{copper}	400 W/mK
h	100W/m ² K
V_f	200mm ³

As shown in Figure 12, for the optimal pin fin, of diameter 2.1mm, a heat flux of 0.8184W is obtained. To dissipate a heat load of 60W, an array of 74 fins would be required, discounting any heat dissipated at the surface area of the heat plate.

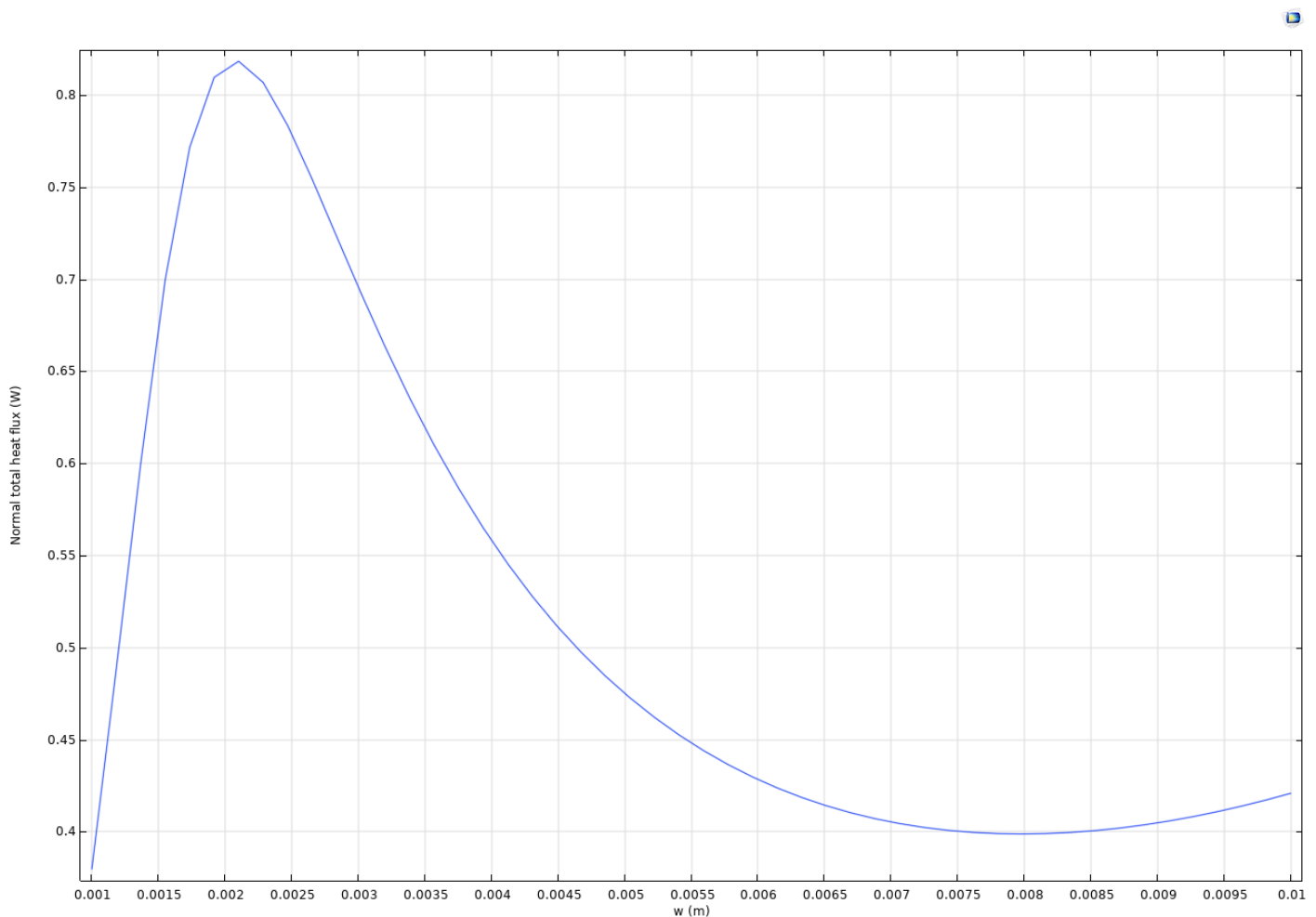


Figure 12: Variation Total Normal Heat Flux of Pin Fin Diameter (width).

The optimal fin could feasibly be manufactured from a copper alloy, however, if higher mechanical strength is required, it may be appropriate to increase thickness. This would increase the number of fins required.

B3: Design Process of a Pin Fin Array for CPU Cooling

For this process, 3D analysis of an array of pin fins attached to a heat spreader base plate was performed. Heat was applied via a small area of 0.5cm^2 , to simulate the heat load through a CPU chip.

Initially, the plate radius was set at the maximum design value of 35mm, for maximum heat distribution. A heat spreader plate thickness of 3mm was selected for structural strength. The weight of the CPU heat sink being concentrated in the base also allows for stability and easy installation and promotes more even thermal contact pressure. This plate, combined with an array of 130 fins gave a volume maximum temperature of 324.43K, or 51.28°C for a steady state heating load of 60W.

The CPU cooler in this case overperformed, so it was necessary to refine the design for better material usage. Reducing the radius of the plate to 29mm and the number of fins to 72 gave a volume maximum temperature of $336.95\text{K}=63.8^\circ\text{C}$.

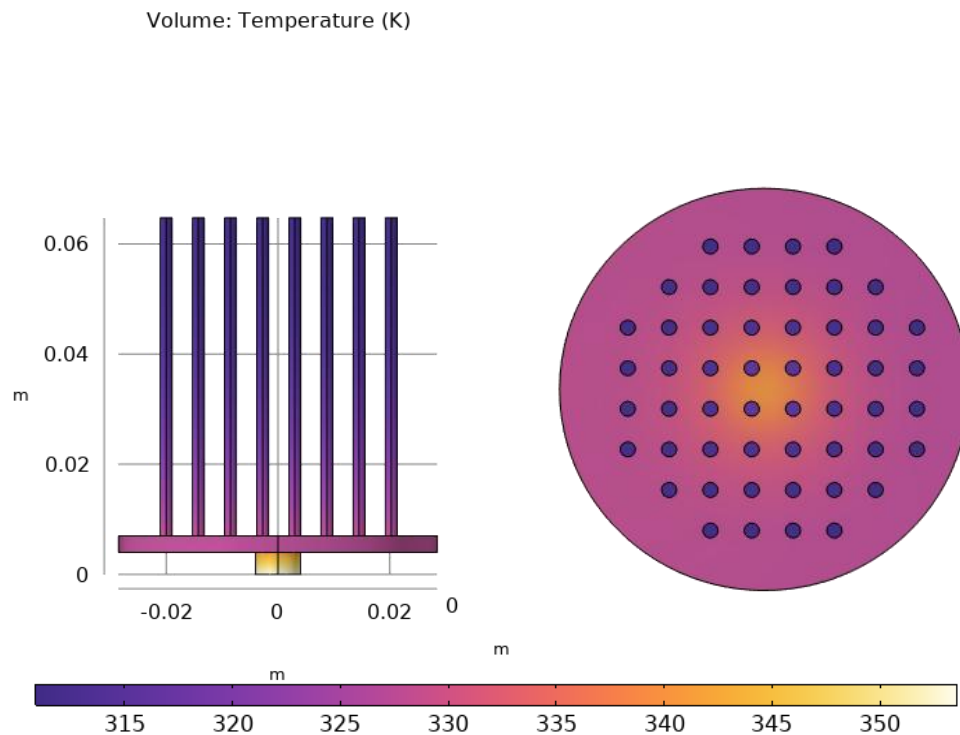


Figure 13: Final Design of Pin Fin with Temperature Surface Plot

The final design (Figure 13) uses an array of 55 pin fins (8×8 minus 9 fins), a plate radius of 28.5mm, a plate thickness of 3mm, and the optimal fin geometry for a 200mm^3 fin as derived in B2. The volume maximum temperature for this heat sink given a 60W heat load over the 0.5cm^2 CPU chip area, is 69.82°C . To reach $T=75^\circ\text{C}$ (348.15K), a constant load of 65W is required (see Figure 14). If overclocking which causes a higher level of heat generation than this is required either a larger array of pin fins, or a better coolant is needed (for example, liquid cooling may be used instead of air cooling).

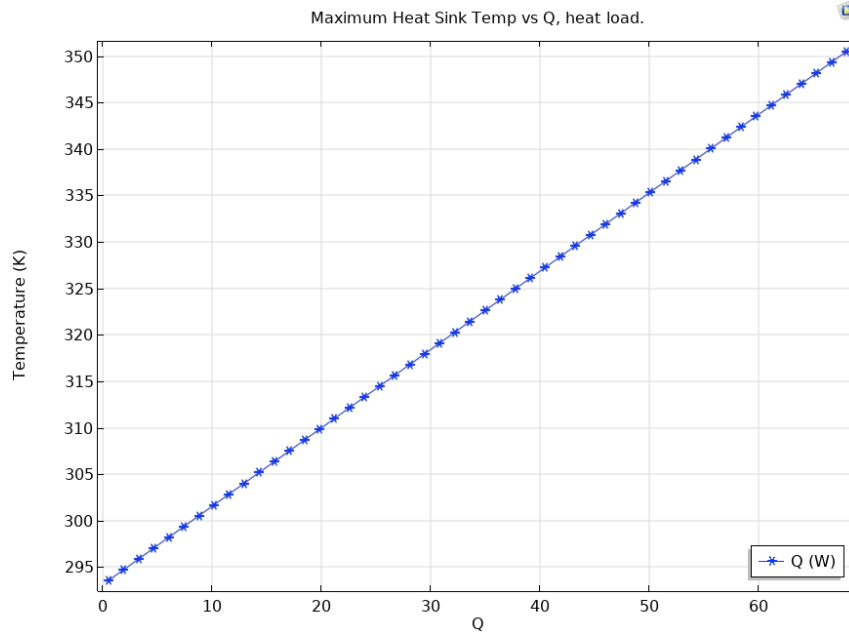


Figure 14: Max Steady State Temp for a range of Q

The required number of fins is slightly lower than predicted in B2. This is likely an effect of the heat spreader: the analysis in B2 assumed 60W was evenly distributed between the pin fins, however in this section a convective boundary condition was applied to the upper surface of the heat spreader. This means that heat transfer through the fins alone is slightly lower than predicted.

B6: Transient Heat Analysis of a Pin-Fin Array

For analysis of the transient behaviour of the heat sink, the "Volume Maximum" function was used in COMSOL to find the highest temperature reached under the applied heat load for times up to 350s.

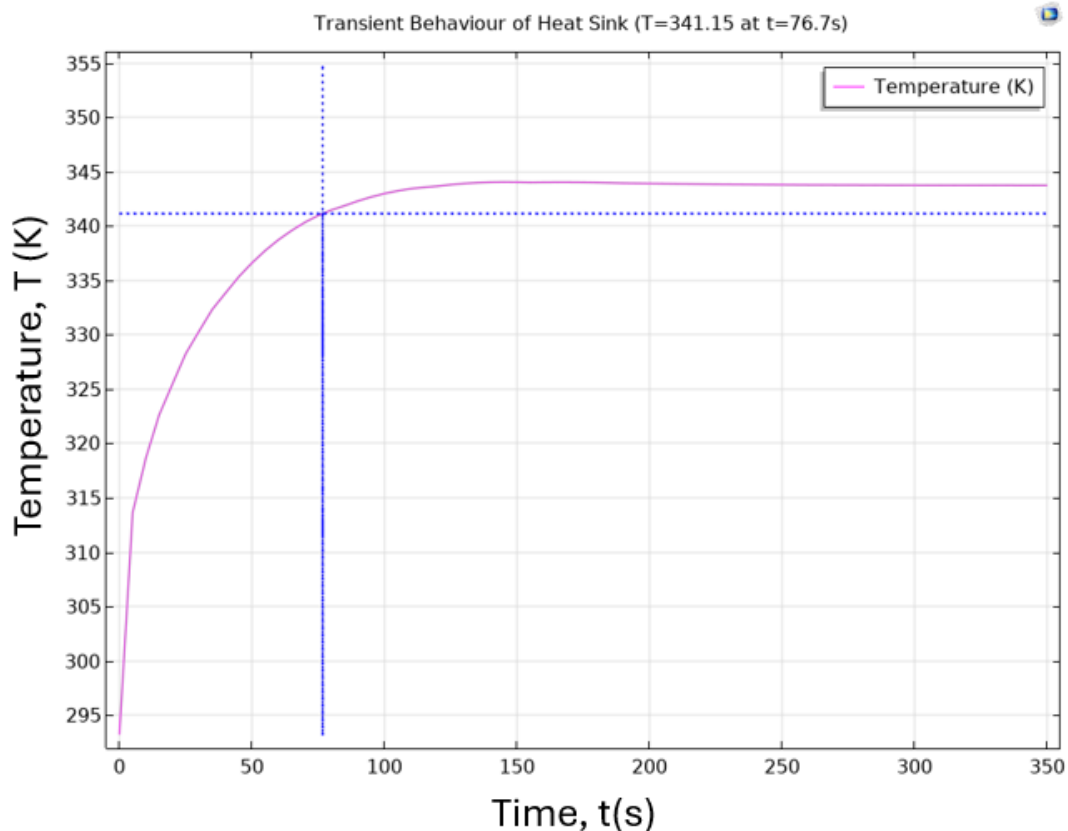


Figure 15: Transient Behaviour of Heat Sink under constant 60W Heating Load. Blue Lines Plotted at T=341.15K, t=76.7s

From figure 15, we can see that under a constant load of $Q=60\text{W}$, the heat sink reaches the temperature limit of 68°C or 341.15K after 76.7s . This shows that there is minimal temperature overshoot, and the heat sink can dissipate a high transient heat impulse without significant temperature increase.

B7: Simulation of CPU Throttling at High Load.

At high CPU loads, for example if the CPU is overclocked for higher performance, it can be necessary to throttle CPU performance to prevent heat damage. In this case, the CPU power is reduced, until temperatures return to the rated range, at which point the CPU may return to full power (overclocked frequency).

For analysis of CPU temperature under a variable heat load, a point probe was placed at the centre of the heat spreader model, in the location where highest temperatures occur. Then, a conditional heat load was assigned to the CPU: when the CPU temperature exceeds the allowed limit, the CPU reduces its clock speed and produces less heat ($Q_{throttle}$). After returning to an allowable temperature, the CPU then returns to its full capacity.

Figure 16 shows the temperature behaviour of the throttled CPU with a lower heat load of $Q=45\text{W}$.

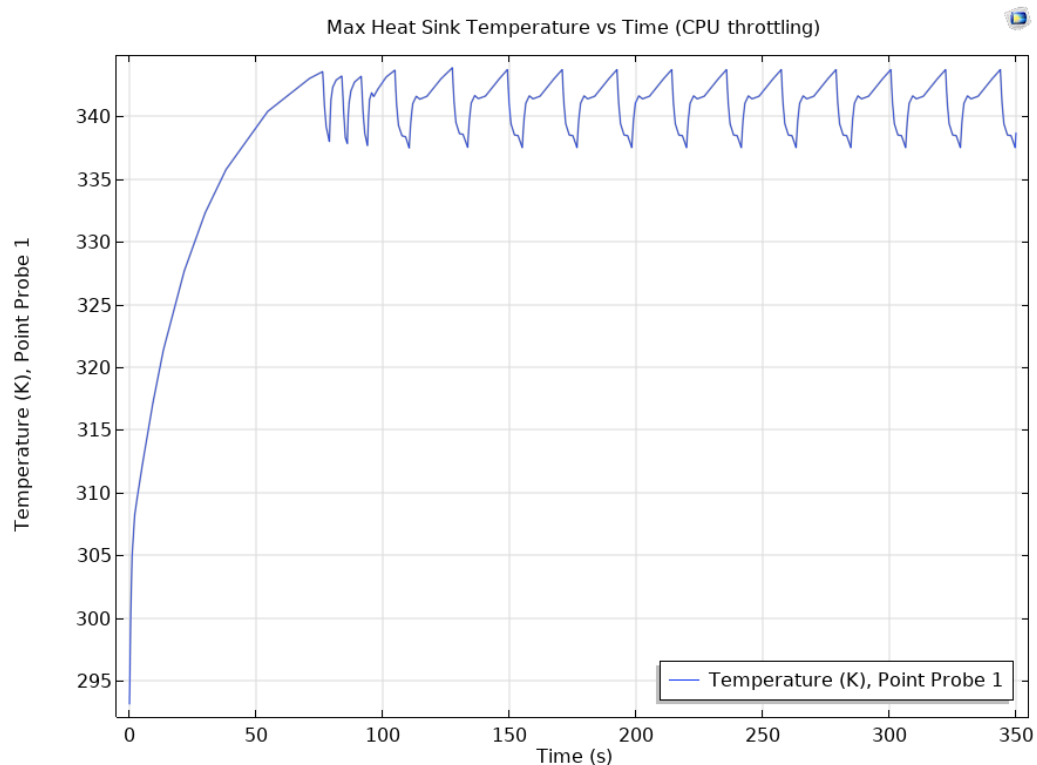


Figure 16: CPU temperature diagram for $Q=45\text{W}$ (throttled value).

Different temperature characteristics may be obtained by varying the gap between $Q_{\text{throttled}}$ and Q_{normal} so that the periodic restriction in power has a different impact. Figure 17 shows the behaviour of the heat sink for $Q_{\text{throttled}}=55\text{W}$.

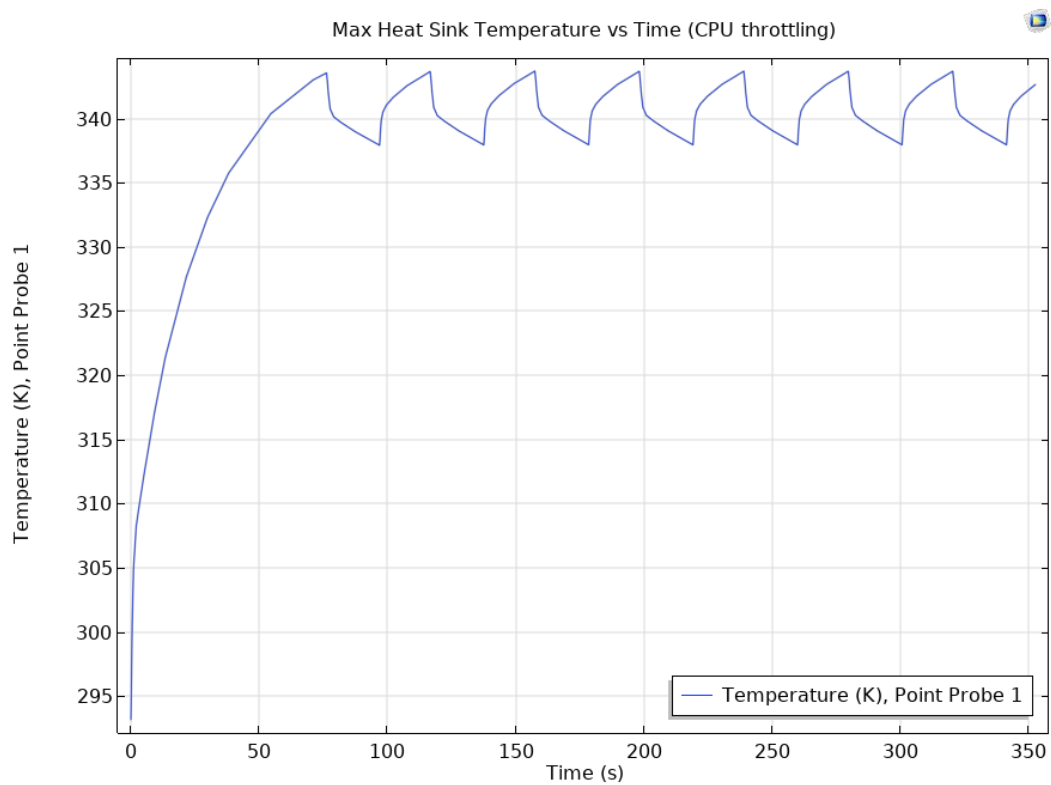


Figure 17: CPU temperature diagram for $Q = 55\text{W}$ (throttled value).

As can be seen above, allowing for a higher value for Q_{throttle} reduces frequency of temperature fluctuations, which may improve CPU and heat sink life. However, this means that the CPU spends more time at the throttled clock frequency, which may negatively impact performance. Another technique, known as “computational sprinting” or “race-to-idle” sees the clock speed maximized for brief periods, followed by an idle period where the processor speed drops to near zero [4]. This allows in some cases for improved performance for better energy efficiency, and lower levels of sustained heating. In figure 18, this effect may be observed- much more time is spent ramping towards the upper CPU temperature limit of 343.15K, which suggests the CPU spends more time at a higher clock frequency, while keeping within the temperature limit. In reality, the switching between idle and active states could be done at a much higher frequency, for further improved average clock speed, however the relationship here is clearly shown. For higher frequency temperature fluctuations and higher level of heat generation, it is important that there is enough thermal resistance to prevent overshoot.

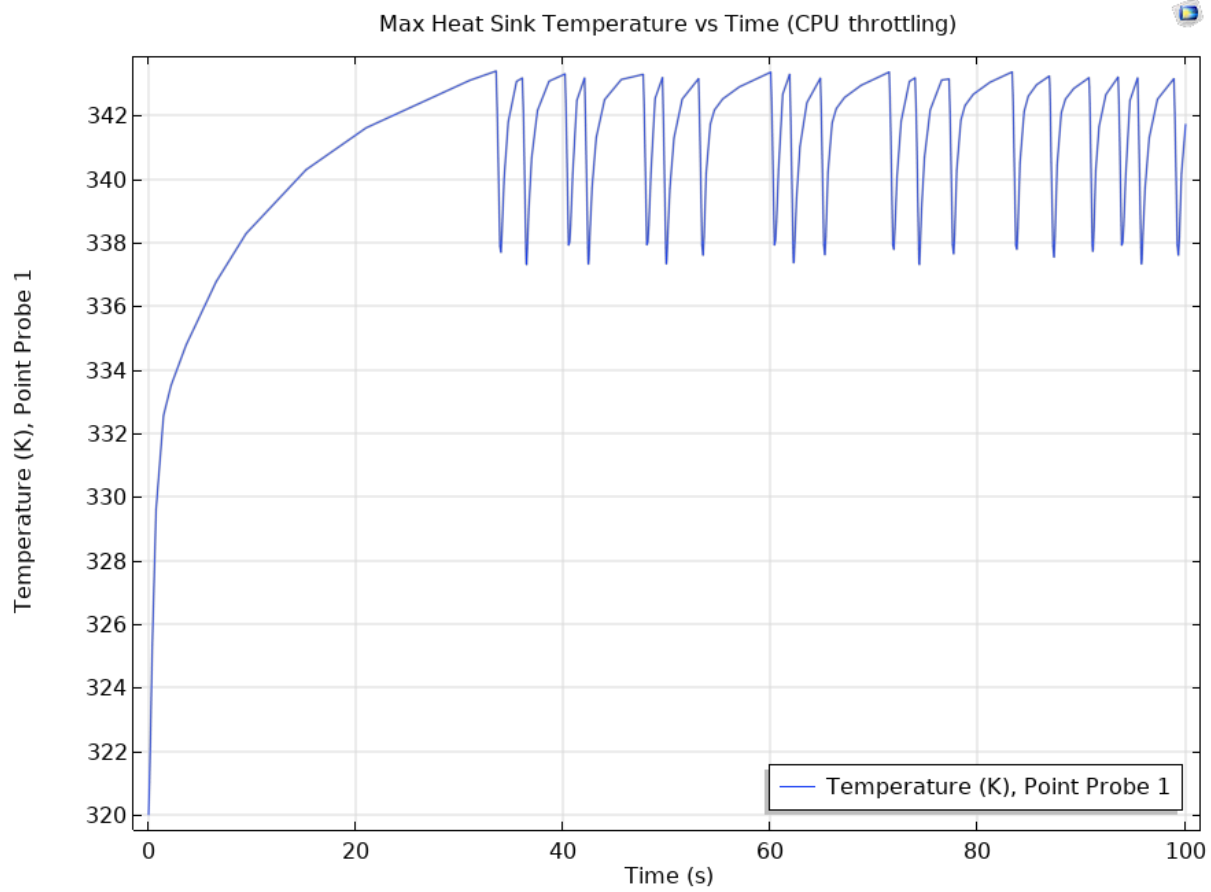


Figure 18: CPU Heat Sink Behaviour for "Race-To-Idle" throttling technique (throttled $Q=0$)

References

- [1] Y. Yang, "SESM3032 Heat Transfer lecture notes, slide 89," *SESM3032 Heat Transfer*, University of Southampton, 2025
- [2] Simply Radiators, "Finned Tube | Low Level Radiator," [Online]. Available: <https://www.simplyradiators.co.uk/finned-tube>. [Accessed: 28-Mar-2025].
- [3] "Nusselt Number," Engineers Edge, [Online]. Available: https://www.engineersedge.com/heat_transfer/nusselt_number_13856.htm. [Accessed: 24-Mar-2025].
- [4] S. Albers and A. Antoniadis, "Race to idle: New algorithms for speed scaling with a sleep state," *ACM Transactions on Algorithms*, vol. 10, no. 3, pp. 1-23, Aug. 2014. [Online]. Available: <https://dl.acm.org/doi/pdf/10.1145/2556953>.